

Anchal Sharma

AI Engineer (LLM Systems) | Text-to-SQL · Retrieval · Evaluation

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Summary

AI Engineer specializing in **LLM systems**, with 3+ years of experience designing and shipping reliable AI for enterprise analytics. Architected and shipped a production **conversational BI (Text-to-SQL) engine** end to end, in which the model produces a structured plan and a **deterministic compiler** generates secure SQL, with join planning, **row-level security** and **answer verification** that catches hallucinated figures. Built the **evaluation infrastructure** that measures it: a 1,429-question retrieval benchmark with ablation studies and a 568-case SQL correctness suite. Leads junior engineers and owns delivery from agent backend to user interface.

Experience

Software Engineer II (SDE-2) *NucleusTeq Pvt. Ltd.*

Indore, India

Jan 2025 – Present

Cosmo Converse — AI Query Engine and Architecture *Python, FastAPI, MongoDB Atlas, Trino, MCP*

- Led end-to-end development of Cosmo Converse, NuoData's conversational BI engine, owning the **Python** backend and the chat frontend, which answers natural-language business questions with validated SQL, charts and dashboards on **Trino** and is **model-agnostic** across LLM providers.
- Architected a **compiler-first Text-to-SQL** design: the LLM emits a structured query plan and a deterministic compiler generates the SQL, with shortest-path **join planning**, double-counting (fan-out) protection, **row-level security** on every table and SQL safety checks, so the model never writes raw SQL.
- Designed **hybrid retrieval** (MongoDB Atlas keyword and **vector search** with **re-ranking**) over the semantic catalog, served to the agent through an **MCP server** with 7 tools: search, context, retrieve, compile, execute, visualize and capabilities.
- Built an **answer verification** layer that validates and auto-repairs every chart and recalculates every number quoted in the answer from result rows, catching **hallucinated figures** before they reach users.
- Designed the **reliability layer**: a 41-code error taxonomy that tells every layer whether a retry can help, **idempotent** persistence, and replay that flags when numbers or metric definitions have changed.

Cosmo Converse — Evaluation Engineering

pytest, MongoDB Atlas, CI

- Built a **retrieval benchmark** over 6 synthetic business domains (488 tables, 10,200 columns, 1,507 relationships, 80 metrics) with 1,429 seeded test questions spanning 39 question types and 48 trap types, locked in **CI** against hand edits.
- Built an **evaluation harness** against the live search endpoints with a waterfall analysis that attributes every miss to a table, column or join and the pipeline step that dropped it; validated the scorer itself by planting known errors.
- Ran **ablation studies** calibrated against the real production catalog and wrote 35 findings reports that identified the top cause of lost results, a missing relevance cutoff and mixed scoring scales, each turned into a proposed fix.
- Built a **568-case SQL correctness suite** across 11 levels (refusals, attack cases, double-counting traps, Spider-style benchmark questions, joins, types, scale) with 567 passing, alongside 834 automated tests overall.
- Built a 31-check live test kit for the **MCP server** covering tool schemas, refusals and end-to-end question-to-chart workflows, and set up specialist **AI review agents** for evaluation and engineering review.

Cosmo — Semantic Layer and Conversational Interface

Next.js, React, ReactFlow, TypeScript

- Built the **semantic domain** modeler that turns tables from any **Trino**-supported source (such as BigQuery and Delta Lake) into a governed, AI-ready **semantic model**: business names, stewardship and certification over read-only catalog metadata, a join-rule editor with draft and published versions, and a readiness check before the agent can query it.
- Built the conversational interface: domain-scoped conversations with follow-up memory, a live trace of every **agent tool step** with timing and token usage, **human-in-the-loop clarification**, typed refusal cards, and freshness controls that re-run a query and confirm whether every figure still matches current data.

Nova — ML and LLM Workbench

Next.js, React, ReactFlow, Module Federation, MLflow

- Led and mentored a team of junior engineers on Nova, NuoData's ML and LLM workbench, owning task breakdown, code reviews and delivery.
- Built the pipeline builder for 20 ML algorithms and 12 **generative AI** task types, and the **RAG** workflow builder (chunking, embedding and vector DB nodes) with **Airflow**-scheduled runs, model catalogs and **MLflow** tracking.

- Led a security and dependency upgrade (Next.js 13 to 15, React 18 to 19, Node 22) that cut critical **CVEs** from 4 to 0 and the **Docker** image from 446 MB to 372 MB.

Software Engineer

July 2023 – Dec 2024

- Built dashboards with **Next.js / React.js, React Flow** and **Plotly.js**, improving pipeline visibility for 5,000+ users.
- Developed **Webpack**-based micro-frontends and backend services (**FastAPI, Node.js**), reducing code duplication by 30%.
- Integrated real-time AI/ML workflows using **JupyterHub** and **Socket.io**, and strengthened security with **Keycloak SSO**.

Software Engineer Trainee NucleusTeq Pvt. Ltd.

Remote

Jan 2023 – June 2023

- Built user registration and blog management features with **React, Java, Spring Boot** and **MongoDB**, with role-based access for admins and employees.
- Maintained code quality with **JUnit 5**, Checkstyle, PMD and SpotBugs.

Software Engineer Intern Vedx India Pvt. Ltd.

Remote

Nov 2022 – Feb 2023

- Built email, SMS and WhatsApp campaign dashboards with **Spring Boot, Next.js** and **MySQL**, including authentication, bulk customer uploads and REST APIs.
- Integrated **Quartz Scheduler, RabbitMQ, AWS, Twilio** and the **WhatsApp Business API**.

Technical Skills

AI Systems and LLM: LLMs, Text-to-SQL, RAG, AI Agents, LangGraph, MCP (Model Context Protocol), Hybrid Search, Vector Search, Embeddings, Re-ranking, Semantic Layer, Human-in-the-Loop, Guardrails, Prompt Engineering, Generative AI

Evaluation and Observability: LLM Evaluation, Retrieval Evaluation, Benchmark Design, Ablation Studies, LangSmith, OpenTelemetry, Prometheus, Grafana, pytest, Jest

Languages: Python, TypeScript, JavaScript, SQL, Java

Backend and Data: FastAPI, Pydantic, Node.js, Spring Boot, REST APIs, Microservices, MongoDB Atlas Vector Search, Trino, BigQuery, Delta Lake, sqlglot, Airflow, MLflow, MySQL

Cloud and DevOps: Docker, AWS (ECR, SSM), CI/CD, SonarQube, Keycloak SSO, Git

Frontend: React.js, Next.js, Redux, TanStack Query, ReactFlow, Module Federation, Micro-frontends, Plotly.js

Education

Maulana Azad National Institute of Technology (MANIT), Bhopal

2020 – 2023

Master of Computer Applications (MCA)

Barkatullah University, Bhopal

2017 – 2020

B.Sc. in Computer Science